

Question	Marking Guidance	Mark	Comments
04.1	(Volume) Correct answer of $57.9 \mu\text{m}^3$ = 2 marks;; OR If volume incorrect, evidence of 2.35-2.45 (as radius) = 1 mark (Times larger) 8 (times larger) (or ECF) = 1 mark;	3 (3 x AO2)	58 = 2 marks 57.91 = 2 marks Accept 54.3/54 = 2 marks Accept 61.6/62 = 2 marks Allow 7.5 – 8.52 ECF Allow any alternative for 8 which shows $463 \div$ their volume

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04.2	1. Nuclear membrane /nucleolus /vesicles/ lysosomes/ribosomes distinct/visible; 2. EM has greater resolution;	2 (1 x AO1, 1 x AO2)	1. Accept invaginations of membrane distinct/visible 1. Reject nucleus 1. Reject mitochondrion 2. assume 'it' refers to electron microscope

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04.3	Stimulating cytotoxic T cells OR Stimulating B cells OR Stimulating phagocytes;	1 (AO1)	Accept 'activate' for stimulating

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04.4	1. Anticodon (on tRNA) binds to (complementary) codon (on mRNA); 2. (tRNA) brings/carries specific amino acid (to	2 (2 x AO1)	

	ribosome);		
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